Alex Fernsby

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Data engineer with seven years in retail and logistics, building batch and streaming pipelines for operational supply-chain data and the warehouses they feed. I own a 40-million-events-per-day shipment pipeline end to end, from Kafka through Airflow into PostgreSQL and BigQuery. I have cut rebuild times, warehouse costs and failed loads with dbt, partitioning and data-quality tests. I want to own a pipeline inside a product team, where logistics data shapes the product and the pipeline owner sits with the people who use it.

Core Competencies

- **Data engineering**: Python, SQL, Apache Airflow, dbt, Kafka, PostgreSQL, BigQuery, Docker, Looker
- **Logistics and order data**: shipment events, delivery-performance mart, customer-orders star schema, partner price feeds
- **Warehouse architecture and modelling**: star schemas, incremental dbt models, partitioning and clustering, warehouse cost control
- **DataOps and CI/CD**: Terraform (Infrastructure-as-Code), GitHub Actions, Docker, Git-based deployments (GitOps-style)
- **Data quality and governance**: Great Expectations, data contracts, on-call runbook, 15-minute feed alerting
- **Stakeholder management and communication**: mentoring, data-modelling reviews, analyst training, cross-functional data contracts

Professional Experience

Tidewater Logistics — **Copenhagen**
Senior Data Engineer — *2022-Present*

- Owns the shipment-events pipeline: 40 million events per day from Kafka into a PostgreSQL and BigQuery warehouse, orchestrated with Apache Airflow.
- Cut the nightly rebuild of the delivery-performance mart from 6 hours to 55 minutes with incremental dbt models.
- Built a Looker explore for the operations team so planners answer delivery-delay questions themselves instead of filing tickets.
- Defined the shipment-events data contract with five producing teams and wrote the on-call runbook for the pipeline.
- Introduced data-quality tests with Great Expectations on 30 core tables; failed loads dropped 70 percent in the first quarter.
- Moved the team's deployments to GitHub Actions and Terraform, so a new environment takes 20 minutes instead of two days.
- Cut the BigQuery bill for the events warehouse by 35 percent by partitioning and clustering the five largest tables.
- Mentors two junior engineers and leads the weekly data-modelling review.
- Ran the interviews for three data-engineering hires and wrote the team's take-home exercise.

Marlow Retail Group — **Utrecht**
Data Engineer — *2018-2022*

- Built the company's first Airflow deployment and migrated 45 cron jobs onto it.
- Added alerting on the 45 Airflow jobs so late partner feeds are noticed within 15 minutes instead of the next morning.
- Wrote Python ingestion services for partner price feeds over REST and SFTP, handling about 2 million rows a day.
- Designed a customer-orders star schema in PostgreSQL used by 12 analysts.
- Trained 12 analysts on dbt and reviewed their first 40 models.
- Supported the move of the warehouse from on-premise PostgreSQL to BigQuery, validating 60 tables row by row.
- Wrote the migration guide for moving reports off the old warehouse, used by four teams.

- **Delft Energy Cooperative** **Delft**
  Junior Analyst *2016-2018*
  - Automated monthly consumption reports with Python and SQL, saving the team about 3 days each month.
  - Built a small Flask dashboard that 25 cooperative members used to read their own consumption.
  - Cleaned five years of meter readings with Pandas and documented the 14 data-quality rules that came out of it.
  - Presented the consumption findings to the cooperative's board twice a year.

Education

- **Delft University of Technology**
  MSc Computer Science (data systems) *2016*
- **Delft University of Technology**
  BSc Computer Science *2014*

Projects

- Open-source dbt package for freight-rate data tests, 120 GitHub stars, github.com/afernsby-example/freight-tests
- Talk "Data contracts in practice", Rotterdam Data Meetup, 2024

Certifications

- Google Cloud Professional Data Engineer, 2023
- dbt Analytics Engineering Certification, 2022

Languages

- English (C1), Dutch (native), German (A2).

Publications

- "Testing freight data with dbt", Data Engineering Weekly guest post, 2024

Honors and Awards

- **Engineering Excellence Award** – Tidewater Logistics (2023).

References

- Available upon request.